Harmonic Sources' Location and Emission Estimation in Underdetermined Measurement System

Fangwei Xu, *Member, IEEE*, Chuan Wang, Kai Guo, Qin Shu, *Member, IEEE*, Zhiquan Ma, and Hongru Zheng

***Abstract*— This article proposes a novel method for estimating multiharmonic sources in the underdetermined measurement system. In the proposed methodology, supposing that the number of harmonic measurements (HMs) is less than that of unknown state variables, the HM system can be mathematically described as underdetermined matrix equations, which has infinitely many solutions corresponding to all possibilities of the locations and the current emissions of the actual harmonic sources. Because the harmonic current emissions from nonharmonic source buses are approximately zero, by fitting the harmonic current emission of every possible nonharmonic source bus combinations and selecting the minimum fitting residual of all combination cases, thus, the true number, locations, and current emissions of harmonic sources are determined. Compared with the existing methods, the proposed method has less dependence on the placement of HMs and requires less prior information of harmonic sources. The effectiveness of the proposed method is verified by simulation and field case, with respect to a high correct location rate of more than 90%.**

***Index Terms*— Genetic algorithm, harmonic sources' location, harmonic state estimation, minimum norm, power system, underdetermined system.**

I. Introduction

NOWADAYS, with a large number of power electronic devices and renewable energy connected to the power grid, harmonic pollution has become an important concern [1]–[6]. Propagation of harmonics in the power system may cause malfunction of relay protection devices and automatic devices, interfere with the communication system [7], and cause serious economic losses [8]. In order to suppress the propagation of harmonics, it is important to locate and estimate harmonic sources in the system.

Harmonic state estimation is the inverse process of harmonic power flow [9], which can effectively estimate harmonic voltage and current of each bus and branch in power grid by a few precisely synchronized harmonic measurements (HMs), without directly monitoring every bus and branch of the system [9]–[14].

Heydt [9] first proposed the least-squares method-based harmonic state estimation to locate the true harmonic sources among multiple suspicious harmonic sources, i.e., the aggregated source at a bus where we are not sure if it contains harmonic loads. Then, the dynamic methods based on the Kalman filter [10], [11] have been used for harmonic injection tracking. In [12], the harmonic source location approach based on the direction of harmonic power flow is used to categorize and rank the suspected buses. In [13], multiple harmonic sources are located by means of an intelligent approach based on fuzzy logic. For taking the uncertainty of network parameters into consideration, the adjusted weight least-squares method is utilized for harmonic state estimation [14]. In order to avoid an ill-conditioned measurement matrix and improve the effectiveness of harmonic state estimation, the optimal procedures for placing HMs are given in [15]–[17].

The above harmonic state estimation methods (usually designed for transmission system [18]) require that the measurement system is overdetermined or critically determined, i.e., the number of synchronized HMs is more than or equal to that of unknown state variables. Due to the cost of HM instruments [14]–[17] or probable instrument failure, the number of synchronized HMs is usually far less than that of unknown state variables, especially for distribution grids [18], which has greatly limited the practicability of the aforementioned methods.

In view of this shortcoming, the harmonic source estimation is proposed in [18]–[20] for the distribution system. Compared with the harmonic state estimation [9]–[14] that observes the whole network harmonic state, the harmonic source estimation only aims at finding the harmonic sources [18] and requires less synchronized HMs, and thus, it is applicable to the underdetermined measurement system, i.e., the number of synchronized HMs is less than that of unknown state variables. In [18]–[20], based on the rough range of harmonic current emissions of the suspicious harmonic sources, the exact harmonic current emissions of the suspicious harmonic sources are estimated by the Bayesian theorem. In particular, in [20], the Metropolis–Hastings approach is employed to estimate the posterior distribution of the suspicious harmonic sources,

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providing a full uncertainty description of estimation results. Subsequently, based on the assumption that the significant harmonic sources are sparsely distributed in the distribution system, the method of harmonic source location and estimation based on compressive sensing is proposed in [1], [2], and [21]; the measurement uncertainties have been further considered in [1] and [2], which can improve the robustness of the methods when the susceptibility of the location results to the placement of HMs is ignored.

In addition, singular value decomposition (SVD) [22]–[26], independent component analysis (ICA) [27], and linear programming [28] are employed to locate the harmonic sources in the underdetermined measurement system. The SVD method might incur a wrong solution due to the infinite set of possible solutions. The ICA method needs the historical harmonic data to reorder and scale the estimated sources. The effectiveness of the linear programming method depends on the placement of HMs. Thus, the practicability of these methods is not satisfactory.

In this regard, this article proposes a novel method for locating and estimating harmonic sources in the underdetermined measurement system. The proposed method is applicable for both transmission and distribution systems under the assumption that the number of harmonic sources is less than that of synchronized HMs. Compared with the existing methods [1], [20] (see Section IV), the proposed method is less dependent on the placement of HMs and has a higher accuracy for estimating harmonic sources. Moreover, the proposed method requires less prior information on harmonic sources.

The proposed methodology is mainly conducted with the following processes: 1) convert the underdetermined measurement problem into solving underdetermined equations; 2) constitute the general solution of the equations with the sum of a particular solution and homogeneous solutions; 3) take the minimum two-norm solution of the equations as the particular solution; 4) for each possible combination of suspicious harmonic sources, determine the coefficient of a homogeneous solution, and then solve the minimum fitting residual of the estimated harmonic current emissions from nonharmonic source buses by a simplified genetic algorithm [29]; and 5) by comparing each minimum fitting residual, all harmonic sources are located and estimated. It is noteworthy that, due to the computational cost reason, the proposed method is applicable only to the networks with a few tens of nodes when the method is directly used.

This article is organized as follows. Section II introduces the analyzed underdetermined measurement system for harmonic sources' location and estimation. Section III describes the proposed algorithm for harmonic sources' location and estimation. Section IV verifies the effectiveness of the proposed method through simulation in the IEEE 14-bus system. Section V verifies the effectiveness of the proposed method through field case. Section VI concludes this article.

II. Underdetermined Measurement System

Consider the h-order HM model [28]

$$\boldsymbol{z}_h = \boldsymbol{H}_h \boldsymbol{x}_h \tag{1}$$

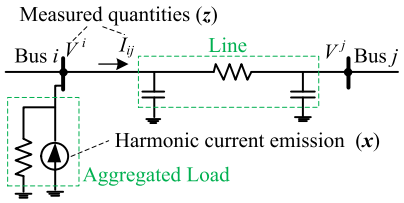


Fig. 1. Example for explaining $\boldsymbol{x}$ and $\boldsymbol{z}$.

where $\boldsymbol{z}_h \in \mathbb{C}^m$ is the known measurement vector obtained from the observed bus harmonic voltage and branch harmonic current. In order to ensure the observability of harmonic sources, generally, assume that $m > k$, where k is the variable of the number of harmonic sources. $\boldsymbol{x}_h \in \mathbb{C}^n$ is the unknown state variable vector (see the example in Fig. 1) composed of the harmonic current emissions of the n suspicious harmonic sources, which generates $\boldsymbol{z}_h$ via network $\boldsymbol{H}_h$. $\boldsymbol{H}_h \in \mathbb{C}^{m\times n}$ is the matrix of the transimpedances between $\boldsymbol{z}_h$ and $\boldsymbol{x}_h$ [13]. Note that all the vectors and matrices in (1) are complex. For simplicity, h is omitted later in this article.

There are several discussions on the solution of (1) as follows.

1) When $m > n$ or $m = n$, (1) is overdetermined or critically determined, and if $rank(\boldsymbol{H}) = n$, then $\boldsymbol{x}$ can be estimated by the least-squares method with $\boldsymbol{x} = (\boldsymbol{H}^{\mathrm{H}}\boldsymbol{H})^{-1}\boldsymbol{H}^{\mathrm{H}}\boldsymbol{z}$ [12], where the superscript H denotes the Hermitian operator (conjugate and transpose).
2) When $m < n$, (1) is underdetermined. In fact, due to the cost of HMs [14]–[17] or possible failure of instrument, the number of available HMs may be less than that of suspicious harmonic sources, even in transmission systems. When (1) becomes underdetermined, the matrix, $\boldsymbol{H}^{\mathrm{H}}\boldsymbol{H}$, is not inverse, so it is invalid to solve (1) directly by the least-squares method [9]–[17]. In this regard, the methods proposed in [18]–[28] are used for solving the problem of harmonic state estimation or harmonic source estimation in the underdetermined measurement system.

It is noteworthy that, for the harmonic source location methods using the multiple-point measured phasor data, all measurements on different buses and branches of the system should be precisely synchronized, which is usually achieved by the synchronization of a global positioning system (GPS) clock.

Moreover, there are some differences in phasor measuring between transmission and distribution systems. The lines in distribution systems are very short, and the angle differences of voltage phasors are usually two orders of magnitude smaller than those in transmission systems [30], which requires higher angular resolution. Also, there are more harmonic loads, distributed generations, and energy storage [31] in distribution systems, resulting in more interharmonics and noises [32]. Besides, closer distance with the customer in the distribution system can also lead to a more sophisticated dynamic behavior of the measured signal. All these interferences and dynamic behaviors may influence the accuracy of phasor measurements [30].

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Fortunately, with the rapid development of phasor measurement units (PMUs) in the distribution systems, the above problems could be better addressed [30], [33]. For example, in [34], based on the digital signal processor (DSP) and GPS, an advanced synchronized measurement system measuring harmonic phasors is designed for wide-area distribution systems. In [35], a PMU prototype based on peripheral component interconnect (PCI) extensions for instrumentation (PXI) modular hardware is proposed for the precise evaluation of harmonic phasors in three-phase distribution systems. Furthermore, the PMU prototypes capable of measuring harmonic phasors in distribution systems have been successfully developed in China [33], which can provide synchronized high-resolution harmonic phasor data, and the updating rate has been up to one cycle (0.02 s) or even faster [33].

In this article, we assume that HMs in the networks can provide synchronized harmonic phasors, which are precisely synchronized and referenced to a common time base, and the measurement accuracy meets the accuracy requirements of Class I instruments reported in standard the International Electrotechnical Commission (IEC) 61000–4–7 [36].

III. Proposed Algorithm for Harmonic Source Location and Emission Estimation

In this section, the proposed algorithm for harmonic source location and emission estimation is given, and each estimation process is concerned with only one harmonic order.

A. Algorithm for Harmonic Source Location and Estimation

When (1) is underdetermined, it has infinite solutions, and its general solution can be expressed as the sum of a particular solution and homogeneous solution.

The particular solution can be derived from the minimum norm solution of (1), whose objective function is

$$\min_{\boldsymbol{x}}\left\{\|\boldsymbol{x}\|_2+\boldsymbol{\lambda}^{\mathrm{H}}(\boldsymbol{z}-\boldsymbol{H}\boldsymbol{x})+(\boldsymbol{z}-\boldsymbol{H}\boldsymbol{x})^{\mathrm{H}}\boldsymbol{\lambda}\right\} \tag{2}$$

where $||\cdot||_2$ denotes two-norm, $\boldsymbol{\lambda}$ is a Lagrange multiplier vector, $\hat{\boldsymbol{x}}_0$ is the minimum two-norm solution satisfying (1), and the particular solution is

$$\hat{\boldsymbol{x}}_0=\boldsymbol{H}^{\mathrm{H}}\left(\boldsymbol{H}\boldsymbol{H}^{\mathrm{H}}\right)^{-1}\boldsymbol{z}. \tag{3}$$

Also, the homogeneous solution of $\boldsymbol{H}\boldsymbol{x}=\vec{\boldsymbol{0}}$ can be expressed as

$$\bar{\boldsymbol{x}}=\sum_{i=\mathbf{1}}^{p}\alpha_i\boldsymbol{\xi}_i \tag{4}$$

where $\bar{\boldsymbol{x}}$ is the homogeneous solution, $\boldsymbol{\xi}_i$ is an n-dimensional column vector, and $\boldsymbol{\xi}_1,\ldots,\boldsymbol{\xi}_p$ are the linearly independent vectors of a homogeneous solution, and α_i is the coefficient of ξ_i. Assume that $\boldsymbol{H}$ has full row rank, so $m=rank(\boldsymbol{H})$, and $p=n-m$. Then, the vectors $\boldsymbol{\xi}_1,\ldots,\boldsymbol{\xi}_p$ can be orthogonalized by the Gram–Schmidt method to reduce the computation time [37].

The general solution of (1) can be obtained from (3) and (4)

$$\hat{\boldsymbol{x}}=\hat{\boldsymbol{x}}_0+\bar{\boldsymbol{x}}=\hat{\boldsymbol{x}}_0+\sum_{i=1}^{p}\alpha_i\boldsymbol{\xi}_i. \tag{5}$$

Set $\boldsymbol{U}=[\boldsymbol{\xi}_1,\boldsymbol{\xi}_2,\ldots,\boldsymbol{\xi}_p]$, and $\boldsymbol{\alpha}=[\alpha_1,\alpha_2,\ldots,\alpha_p]^{\mathrm{T}}$, where the superscript T denotes transpose. Equation (5) can also be expressed as

$$\hat{\boldsymbol{x}}=\hat{\boldsymbol{x}}_0+\boldsymbol{U}\boldsymbol{\alpha} \tag{6}$$

where the coefficient vector $\boldsymbol{\alpha}$ is unknown.

Next, it presents how to determine the number, locations, and current emissions of harmonic sources by choosing coefficient $\boldsymbol{\alpha}$. Assume that there are k harmonic sources in the n-dimensional vector $\boldsymbol{x}$, so there are $(n-k)$ buses without harmonic sources, and their harmonic current emissions can be considered to be approximately zero. Accordingly, there are $(n-k)$ zero elements in $\hat{\boldsymbol{x}}$. Extracting the $(n-k)$ zero elements of $\hat{\boldsymbol{x}}$ from (6), we have the fitting equations

$$\hat{\boldsymbol{x}}_0^{n-k}\approx-\boldsymbol{U}^{n-k}\boldsymbol{\alpha} \tag{7}$$

where $\hat{\boldsymbol{x}}_0^{n-k}$ and $\boldsymbol{U}^{n-k}$ are $(n-k)$ dimensional subvectors of $\hat{\boldsymbol{x}}_0$ and $(n-k)\times p$ order submatrix of $\boldsymbol{U}$, respectively. It is obvious that (7) contains $(n-k)$ equations and p unknown variables. As mentioned before, $p=n-m$ and $k<m$, so $n-k>p$; hence, (7) is an overdetermined equation and can be solved by the least-squares method. We have

$$\hat{\boldsymbol{\alpha}}=-\left(\left(\boldsymbol{U}^{n-k}\right)^{\mathrm{H}}\boldsymbol{U}^{n-k}\right)^{-1}\left(\boldsymbol{U}^{n-k}\right)^{\mathrm{H}}\hat{\boldsymbol{x}}_0^{n-k} \tag{8}$$

and the fitting residual vector

$$\hat{\boldsymbol{r}}(k)=\hat{\boldsymbol{x}}_0^{n-k}+\boldsymbol{U}^{n-k}\hat{\boldsymbol{\alpha}}. \tag{9}$$

Note that k harmonic sources have $C_n^k=n!/(k!(n-k)!)$ possible locations in the n suspicious harmonic source buses, where $(\cdot)!$ denotes factorial. Thus, it is necessary to figure out the C_n^k estimation of $\boldsymbol{\alpha}$ and the corresponding residual vector. Since the harmonic current emissions of the buses without harmonic sources are close to zero, the real harmonic sources can be located and estimated by selecting the $(n-k)$ buses combination whose residual vector has the minimum two-norm.

Let

$$r(k)=\|\hat{\boldsymbol{r}}(k)\|_2. \tag{10}$$

The minimum fitting residual is set as $r_{\min}(k)$.

If the true number of harmonic sources k_{true} is given, let $k=k_{\text{true}}$, and use (2)–(10); $r_{\min}(k)$ and the corresponding $\hat{\boldsymbol{\alpha}}$ can be estimated. Then, by substituting $\hat{\boldsymbol{\alpha}}$ into (9), the emissions and locations of the k harmonic sources are determined. However, k_{true} is usually unknown in practice; this article proposes a new empirical criterion for confirming the number of harmonic sources from the possible range $k\in[1,m)$.

By making use of the Monte Carlo method [38], i.e., a large number of simulation studies, we find that, when k grows and closes to the true number of harmonic sources, $r_{\min}(k)$ becomes smaller, and only if k exceeds the true number of harmonic sources, the increment of minimum fitting residual, defined as $\Delta r_{\min}(k)=|r_{\min}(k+1)-r_{\min}(k)|$, is close to zero. Thus, the increment is utilized as the criterion for deciding the number of harmonic sources, and the steps can be summarized as follows:

1) Set $k=1$.
2) Calculate $\Delta r_{\min}(k)=|r_{\min}(k+1)-r_{\min}(k)|$.

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3) If $\Delta r_{\min}(k) < \delta |r_{\min}(1)|$ or $k = m - 1$, this k is judged as the estimated number of harmonic sources k_{est}; otherwise, let $k = k + 1$ and back to step 2, where $\delta = 0.05$ is an empirical value determined by the Monte Carlo method.

After determining k_{est}, through (2)–(10), the harmonic current emissions $\boldsymbol{x}$ of the k_{est} harmonic sources are obtained.

B. Simplified Genetic Algorithm for Reducing Operations

It is obvious that the above processing of the method needs a large amount of computation, especially for a large electric power system with a lot of HMs. Because the genetic algorithm can quickly obtain global solutions of various functions [16], [29], in this section, the standard genetic algorithm [29] is slightly simplified, and then, it is employed to perform the calculations of $r_{\min}(k)$ for reducing the computation of the method. In our simplified version, there is no crossover operation, and the initial population is generated according to the element of minimum norm solution of (2), not randomly.

First, define an n-dimensional row vector whose element is set as 0 or 1; the element index indicates the bus number of suspicious harmonic sources, where the 0-element corresponds to the harmonic source bus and the 1-element corresponds to the nonharmonic source bus. Since the number of harmonic sources is k, and thus, the total number of 0-elements is k, the total number of 1-elements is $(n - k)$. Besides, $1/r(k)$ is chosen as the *fitness function* of the genetic algorithm, N is iterations, and M is the population size.

1) Generate Initial Population: The initial population is chosen according to the minimum norm solution of (2). It is easy to see that the larger the element amplitude of the minimum norm solution vector is, the more possible the corresponding bus has a harmonic source. Based on the magnitude of each element in $\hat{\boldsymbol{x}}_0$, from big to small in principle, k 0-elements, and $(n-k)$ 1-elements are arranged to generate M individuals (*binary coding vector*). This step is further explained by the following example.

Suppose that $M = 3$, $k = 2$, and $n = 5$. When $|\hat{\boldsymbol{x}}_0(3)| > |\hat{\boldsymbol{x}}_0(4)| > |\hat{\boldsymbol{x}}_0(2)| > |\hat{\boldsymbol{x}}_0(1)| > |\hat{\boldsymbol{x}}_0(5)|$ [where $\hat{\boldsymbol{x}}_0(i)$ is the ith element of $\hat{\boldsymbol{x}}_0$ calculated by (2)], bus 3 and bus 4 are most likely the harmonic source buses, and bus 5 is least likely, so the third and fourth elements of the first individual are set to zero; accordingly, the three individuals of the initial population can be described as follows: first: [1 1 0 0 1]; second; [1 0 0 1 1]; and third: [0 1 0 1 1].

2) Fitness Calculation: The *fitness function* $1/r(k)$ is used to calculate the fitness value of each individual in the current population.

3) Selection: According to the fitness value of each individual, the individual with the largest fitness value calculated by step 2 is selected first. Then, the remaining $(M/3 - 1)$ individuals were selected by the Roulette Wheel Selection method [39], where the larger the fitness of each individual is, the higher the probability of being selected is.

4) Mutation: First, the individual with the largest fitness value is directly copied to the next generation, and then, the mutation operation is carried out for each individual in the current population to obtain the remaining $(M - 1)$ individuals. For each individual to be mutated, the mutation operation is first randomly selecting a 0-element and a 1-element of this individual and then changing them into 1 and 0, respectively. For the individual with the largest fitness value, the mutation operation is repeated $(M/3 + 1)$ times to form $(M/3 + 1)$ different individuals. For the remaining $(M/3 - 1)$ individuals, each of them is mutated three times to form three different individuals. After applying the mutation operator, there are a total of M individuals in the current population.

5) Repeat Steps (2)–(4) N Times: If the population at the Nth iteration is the same as that at the $(N - 1)$th iteration, stop iterating. Else, repeat steps (2)–(4) until the latest population is unchanged. Then, the individual with the largest fitness in the current population is selected as the final result, and the corresponding *binary coding vector* represents the locations of the k harmonic sources.

It is noteworthy that, although the genetic algorithm can greatly reduce the operations, the computational time is still long for the networks with hundreds of nodes. Thus, if the proposed method is directly used, it is applicable only to the networks with a few tens of nodes due to the computational cost reason. However, in terms of the radial grids with tens or hundreds of nodes, a hierarchical harmonic source location strategy, i.e., divide the network into multiple areas (as reported in [33]) for locating harmonic sources, can be used to further reduce operations.

The flowchart of the proposed method is given in Fig. 2.

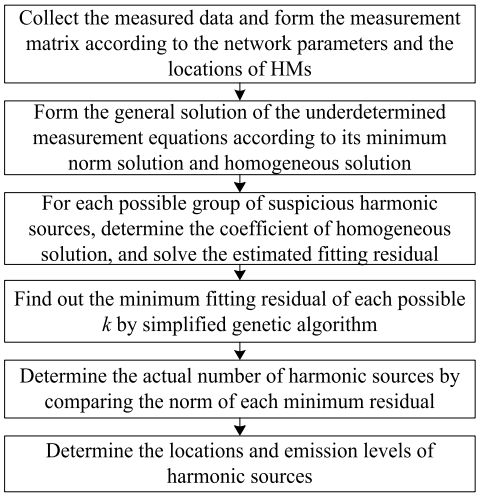


Fig. 2. Flowchart of the proposed method.

IV. Simulation Verification

In the simulation, the proposed method is compared with the linear programming method [28], the block orthogonal matching pursuit (BOMP) method [1], and the Bayesian method [20], where the linear programming method is designed for transmission systems, and the BOMP and Bayesian methods are designed for distribution systems.

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TABLE I
MEASUREMENT ERRORS

Class	Measurement	Conditions	Measurement errors [Max error]
I	Voltage	$U_m \geq 1\%U_n$	$\pm 5\%U_m$
		$U_m < 1\%U_n$	$\pm 0.05\%U_n$
	Current	$I_m \geq 3\%I_n$	$\pm 5\%\%I_m$
		$I_m < 3\%I_n$	$\pm 0.15\%I_n$

TABLE II
PARAMETER ERRORS

Parameter errors [Max error]			
Lines	Transformers	Capacitors	Loads
±5%	±5%	±5%	±20%

Thus, the comparison with the linear programming method is carried out using the IEEE 14-bus system (transmission system), and the comparisons with the BOMP method and the Bayesian method are carried out using the IEEE 13-bus system (distribution system). The two systems are, respectively, the Test Systems No. 1 and No. 2 reported in [40].

The measurement errors of the envisaged instrumentation are shown in Table I, in which the measurement errors of voltage measurements and current measurements are in accordance with the accuracy requirements of Class I instruments reported in standard IEC 61000-4-7 [36]. In addition, for harmonic order h, 1^*h crad phase angle errors (maximum errors) for both voltage measurements and current measurements are taken into account [1]. Furthermore, in Table II, the parameter errors with respect to their nominal values are also considered. The load parameter error is set to $\pm 20\%$ which corresponds to the possible variations of load power [20].

The result uncertainties caused by measurement errors and parameter errors are analyzed by the Monte Carlo method [41]. The Monte Carlo method has been successfully used to solve the problem that the analytical law of uncertainty propagation is either difficult or impossible to apply [42].

For each considered case, $Q = 100$ times of Monte Carlo simulations are performed, in which the measurement errors (in terms of both amplitude and phase angle) and parameter errors are with respect to the Gaussian distribution with zero mean, and its standard deviation equals to one-third of the maximum error. The load active and reactive powers are also extracted from Gaussian distribution within the range [80%, 120%] of their rated values.

Moreover, in order to simulate a real power grid, the small harmonic current has been injected into the buses only with linear loads. The magnitude of the small current is generated by Gaussian distribution with zero means, its standard deviation equals 0.1% of the magnitude of the load current at the fundamental frequency, and its angle is randomly selected between 0° and 360°. The all-measured values are obtained by performing harmonic power flow. Set $N = 20$ and $M = 30$ for the IEEE 14-bus system, and $N = 15$ and $M = 20$ for the IEEE 13-bus system.

The following performance metrics are defined to evaluate the performance of the methods.

A. Correct Location Rate

$$\mathrm{CLR} = \frac{\sum_{i=1}^{Q} k^i}{\sum_{i=1}^{Q} \max\left(k_{\mathrm{true}}, k_{\mathrm{est}}^i\right)} \times 100\% \tag{11}$$

where Q is the simulation times, k_{true} represents the true number of harmonic sources, and k_{est}^i is the estimated number of harmonic sources in the ith simulation and equals to the k_{est} determined by steps 1)–3) in Section III-A. k^i is the number of correctly located harmonic sources among the estimated k_{est}^i harmonic sources in the ith simulation. The correct location rate (CLR) can comprehensively reflect the accuracy of locating the harmonic source in terms of the position and the number of harmonic sources.

B. Correct Estimation Percentage of the Number of Harmonic Sources

$$\mathrm{CNP} = \left(\frac{1}{Q}\sum_{i}^{Q} \kappa^i\right) \times 100\%$$
$$\text{s.t. } \kappa^i = \begin{cases} 1, & k_{\mathrm{est}}^i = k_{\mathrm{true}} \\ 0, & k_{\mathrm{est}}^i \neq k_{\mathrm{true}}. \end{cases} \tag{12}$$

C. Absolute Percentage Error

$$\mathrm{APE} = \left(\left\| \boldsymbol{x}_{\mathrm{est}}^i - \boldsymbol{x}_{\mathrm{true}} \right\|_2 / \left\| \boldsymbol{x}_{\mathrm{true}} \right\|_2\right) \times 100\% \tag{13}$$

where $\boldsymbol{x}_{\mathrm{true}}$ is the vector of true harmonic current emissions of suspicious harmonic sources and $\boldsymbol{x}_{\mathrm{est}}^i$ is the vector of estimated harmonic current emissions in the ith simulation.

The estimation uncertainties are mainly shown in the boxplot of absolute percentage errors (APEs) calculated by Q times of simulations.

D. Root Mean Square Percentage Error

$$\mathrm{RMSPE} = \sqrt{\frac{1}{q}\sum_{i=1}^{q} \left(\mathrm{APE}^i\right)^2} \times 100\% \tag{14}$$

where q is the number of APEs and APE^i represents the ith estimated APE. In this article, the root mean square percentage error (RMSPE) is calculated from the 95% probability values of APEs.

Moreover, it is necessary to define a threshold ϕ for locating harmonic source [1], so as to prevent the wrong location results caused by very small negligible harmonic emissions. Here, if $\hat{\boldsymbol{x}}(i) > \phi_i$, the load is labeled as harmonic load; otherwise, it is termed linear load, where ϕ_i represents the harmonic threshold for the load at bus i. Note that the expected harmonic values and harmonic emission limits between transmission and distribution systems are very different, which is determined by their voltage levels, short-circuit capacities, and load powers [43]. For the sake of simplicity, in this article, ϕ_i is set to 0.5% of its nominal current at the fundamental frequency for each considered harmonic order. ϕ_i defined here is just used for finding the harmonic sources with nonnegligible emissions. In real applications, the actual harmonic emission limit can be determined by actual demands or IEC standards; then, the real significant harmonic sources can be screened out from the located harmonic sources.

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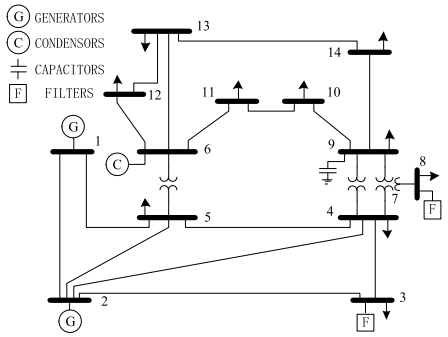


Fig. 3. IEEE 14-bus system.

TABLE III
HARMONIC SOURCE LOCATIONS AND METER PLACEMENT

Case	Harmonic load locations at buses			Meter placement
Case A1	3	10	/	nine meters are randomly
Case A2	3	10	12	placed

TABLE IV
HARMONIC SPECTRUMS [40]

Harmonic order	Percent	Relative Angle (rad)
1	100.00%	0.00
5	19.41%	-0.32
7	13.09%	1.07
11	7.58%	0.74
13	5.86%	2.06

TABLE V
ESTIMATED RESULTS OF CASE A1

	Liner programming method			Proposed method		
Harmonic Order	*CNP*	*CLR*	*RMSPE*	*CNP*	*CLR*	*RMSPE*
5th	13%	40.46%	16.98%	100%	98.5%	3.86%
7th	7%	34.75%	40.80%	99%	97.51%	3.21%
11th	66%	32.11%	90.42%	95%	95.15%	10.26%
13th	12%	21.68%	91.93%	92%	91.90%	15.73%

1) IEEE 14-Bus System: In this section, the simulation is performed on the IEEE 14-bus test system (as shown in Fig. 3) using MATLAB 9.5. The used system is a transmission system, and the network parameters are the same as those reported in [40].

The two cases with different placement of harmonic loads and meters are analyzed in this section and reported in Table III, where the number of harmonic loads in case A1 and case A2 is, respectively, two and three, and the locations of the nine HMs are randomly placed at all possible branches. For each considered harmonic load, the harmonic current spectrums in nominal conditions are given in Table IV. Their harmonic current emissions fluctuate with the changing of load powers.

In case A1, the average computational time of the proposed method is 0.01 s; hence, we can quickly locate harmonic sources in small networks.

The estimated results of case A1 are reported in Table V and Fig. 4. In Table V, the proposed method can well estimate and locate the harmonic source with respect to small RMSPEs, and both correct estimation percentage of the number of harmonic sources (CNP) and CLR for each considered

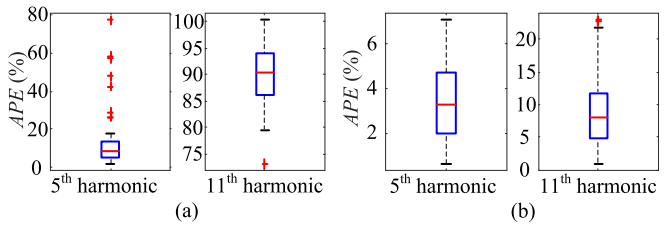


Fig. 4. Estimated APEs (IEEE 14-bus system, case A1). (a) Linear programming method and (b) Proposed method.

harmonic order are larger than 90%. However, due to the relatively large measurement errors on phase angles of 11th and 13th harmonics, the estimation errors from 11th and 13th harmonics are obviously larger than those from fifth and seventh harmonics, while, for the linear programming method, its estimation results correspond to much lower CNP and CLR, and the estimated RMSPEs for 11th and 13th harmonics are above 90%. This is because the estimation performance of the linear programming method is highly dependent on the placement of HMs, and the randomly placed HMs will decrease their effectiveness.

The APEs estimated by the Q times simulations for 5th and 11th harmonics are shown in the boxplots Fig. 4. In the figures, the red line represents the median of the results; the values of lower and upper sides of the blue rectangle are, respectively, first and third quartiles (a_2 and a_1); the two black horizontal lines connected with the rectangle are, respectively, the nonoutlying minimum value [the largest one small than $a_2 - 1.5(a_1–a_2)$] and the maximum value [the smallest one large than $a_1 + 1.5(a_1–a_2)$]; and the red "+" indicates an outlier, which is smaller than the nonoutlying minimum value or larger than the nonoutlying maximum value [44]. Note that the outlier with a large APE usually corresponds to a wrong located result of the harmonic source. In other words, if the harmonic sources are wrong located, the corresponding APE may be very large.

According to the APEs in Fig. 4(a), the APEs estimated by the linear programming method are very large especially for the 11th harmonic, which is concentrated at 90%. In comparison, in Fig. 4(b), most of the APEs estimated by the proposed method are concentrated on low error levels (below 15%), and there are only a few outliers.

In case A2, the number of harmonic sources is increased to three to further verify the proposed method, and the meter placement has no change. In this case, the average computational time of the proposed method is 0.02 s. Table VI and Fig. 5 report the estimation results. According to the results in Table VI, for the proposed method, all the estimated CNP and CLR are larger than 94%, and the largest RMSPEs is below 15%, while, for the linear programming method, the estimated CNP and CLR are lower than 50%, and the smallest RMSPE is still larger than 19%.

Fig. 5 shows the estimated APEs. As for the linear programming method, most APEs in Fig. 5(a) are greater than 20% for the seventh harmonic and 90% for the 13th harmonic. In comparison, in Fig. 5(b), the APEs estimated by the proposed method are still focused at low error levels that are,

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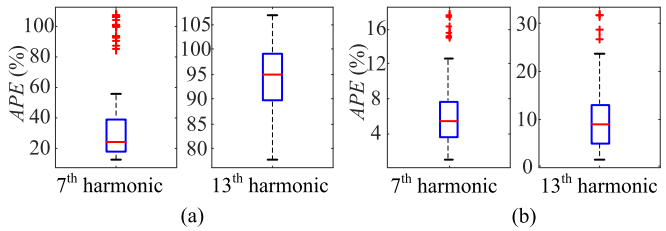


Fig. 5. Estimated APEs (IEEE 14-bus system, case A2). (a) Linear programming method and (b) Proposed method.

TABLE VI
ESTIMATED RESULTS OF CASE A2

Harmonic Order	Liner programming method			Proposed method		
	CNP	*CLR*	*RMSPE*	*CNP*	*CLR*	*RMSPE*
5th	8%	46.86%	19.92%	100%	98%	3.76%
7th	7%	42.58%	48.78%	98%	96.33%	8.11%
11th	51%	45.99%	95.51%	98%	94.37%	14.11%
13th	28%	38.28%	95.07%	99%	94.82%	12.37%

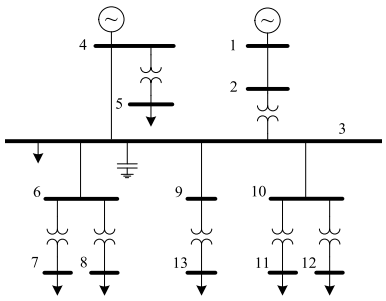


Fig. 6. IEEE 13-bus system.

TABLE VII
HARMONIC SOURCE LOCATIONS AND METER PLACEMENT

Case	Harmonic load locations at buses			Meter placement
Case B1	7	11	/	six meters are randomly placed
Case B2	7	11	12	

TABLE VIII
HARMONIC SPECTRUMS [40]

Harmonic order	Percent	Relative Angle (rad)
1	100.00%	0.00
5	18.24%	-0.97
7	11.90%	-1.47
11	5.73%	-2.51
13	4.01%	-3.06

respectively, 5% and 10% for the seventh and 13th harmonics. These results indicate that the effectiveness of the proposed method is less dependent on the placement of HMs.

2) IEEE 13-Bus System: In this section, the simulation is performed on the IEEE 13-bus test system (as shown in Fig. 6), a radial distribution test system reported in [40]. The two cases reported in Table VII are discussed here, where the harmonic current spectrums of harmonic loads are presented in Table VIII. In each considered case, six HMs are randomly selected from branches 3–6, 3–9, 3–10, 6–7, 6–8, 10–11, and 10–12.

The assumptions or prior information of the BOMP method, the Bayesian method, and the proposed method are presented

TABLE IX
ASSUMPTIONS OR PRIOR INFORMATION

BOMP method	Bayesian method	Proposed method
known k_{true}, and assume $k_{true}<m$	known the range of harmonic current emissions of suspicious harmonic sources	assume $k_{true}<m$

TABLE X
EXAMPLE OF THE DATA CHARACTERIZING THE HARMONIC LOADS

Harmonic order	Harmonic load locations at buses					
	7		11		12	
	Amp. (p.u.)	Ang. (rad)	Amp. (p.u.)	Ang. (rad)	Amp. (p.u.)	Ang. (rad)
5	0.0221	-1.30	0.0092	-1.75	0.0684	-1.78
7	0.0144	-1.80	0.0060	-2.25	0.0446	-2.27
11	0.0069	-2.84	0.0029	3.00	0.0215	2.97
13	0.0049	2.89	0.0020	2.44	0.0150	2.41

TABLE XI
ESTIMATED RESULTS OF CASE B1

Harmonic Order	BOMP method			Proposed method		
	CNP	*CLR*	*RMSPE*	*CNP*	*CLR*	*RMSPE*
5th	100%	98.5%	2.34%	100%	100%	2.22%
7th	100%	93.5%	23.92%	100%	100%	2.91%
11th	100%	100%	4.22%	100%	99.5%	4.22%
13th	100%	94.5%	25.20%	99%	99.5%	5.47%

in Table IX, where a rough prior knowledge of the current emissions of suspicious harmonic sources is necessary for the Bayesian method [20]. The detailed prior information configuration of the Bayesian method is given as follows.

Step 1: For the suspicious harmonic source that is harmonic load, the prior knowledge of its injected current (in terms of both real and imaginary parts) has been represented by means of uniform distributions in the range of $[x_{\mathrm{MINi}}, x_{\mathrm{MAXi}}]$, where i is the bus No. of harmonic loads, $x_{\mathrm{MINi}} = 0$, and x_{MAXi} is twice the value of the harmonic current shown in Table X. Here, the prior information configuration of the harmonic load is similar with that in [20].

Step 2: For the suspicious harmonic source that is linear load, the prior knowledge of its injected current (in terms of both real part and imaginary part) has been represented by means of uniform distributions within the range $[0, \rho I_{\mathrm{Ni}}]$, where I_{Ni} is the amplitude of the nominal fundamental current of the aggregated load at bus i. In order to figure out the influence of prior information of linear load (which is not discussed in [20]), a further discussion is necessary, i.e., changing the value of ρ within [0%, 15%] to show the performance of the method.

Table X shows an example of the data characterizing the harmonic loads and the amplitude of the main harmonic currents injected by the harmonic loads in the network when all nodes work in their nominal conditions of case B2. As, for case B1, only the loads at bus 7 and 11 are harmonic loads, the load at bus 12 works as a linear load, and the prior knowledge of its injected current is assigned based on Step-2.

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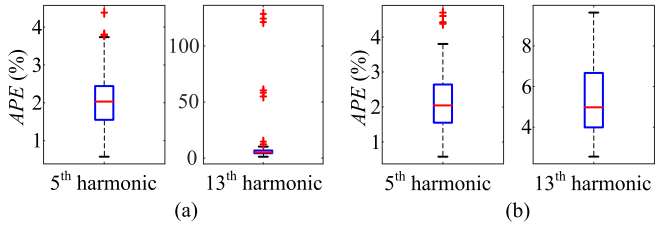


Fig. 7. Estimated APEs (IEEE 13-bus system, case B1). (a) BOMP method and (b) Proposed method.

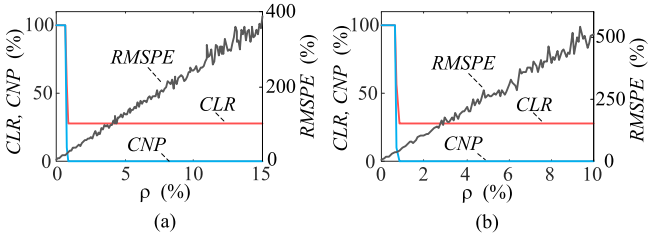


Fig. 8. Results of the Bayesian method (IEEE 13-bus system, case B1). (a) 7th harmonic, Bayesian method and (b) 11th harmonic, Bayesian method.

TABLE XII
ESTIMATED RESULTS OF CASE B2

Harmonic Order	BOMP method			Proposed method		
	CNP	*CLR*	*RMSPE*	*CNP*	*CLR*	*RMSPE*
5th	100%	92.33%	40.73%	100%	98.67%	2.53%
7th	100%	100%	3.15%	100%	97.33%	3.34%
11th	100%	96%	9.67%	100%	99.33%	4.28%
13th	100%	99.67%	5.60%	99%	98.67%	6.51%

In case B1, the average computational time of the proposed method is 0.01 s. Table XI and Figs. 7 and 8 show the estimated results of case B1. According to the CNP and CLR in Table XI, both the BOMP method and the proposed method can locate harmonic sources with high accuracy, i.e., all CLRs for considered harmonic orders are greater than 90%. While the estimated RMSPEs of the BOMP method are relatively large, exceed 20% for the seventh and 13th harmonics. Note that, because of the prior knowledge of the number of harmonic sources (i.e., $k_{\text{est}} = k_{\text{true}}$), the estimated CNP of the BOMP method is always 100%.

In Fig. 7, the estimated APEs correspond to a low estimation error except for the 13th harmonic of the BOMP method. It is possible to see that the outliers (red "+") with large APEs will increase the corresponding RMSPE.

Fig. 8 shows the estimation results of the Bayesian method under different ρ's, where the curves of CLR, CNP, and RMSPE relative to ρ are, respectively, shown. According to the curves, when $\rho = 0$, the Bayesian method shows a good estimation result, and both CLR and CNP equal to 100%. However, when ρ is increased to exceed 1%, the estimation results become worse. The estimated RMSPEs increase significantly with the increase in ρ. These results confirm that the performances of the Bayesian method are highly dominated by the prior information of suspicious harmonic sources that are actually linear loads.

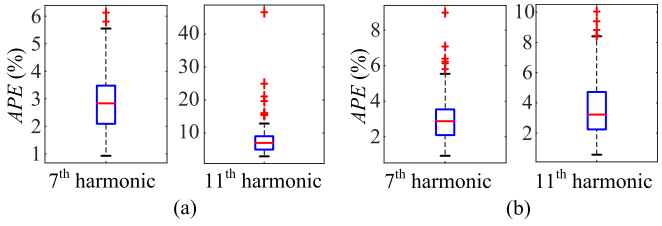


Fig. 9. Estimated APEs (IEEE 13-bus system, case B2). (a) BOMP method and (b) Proposed method.

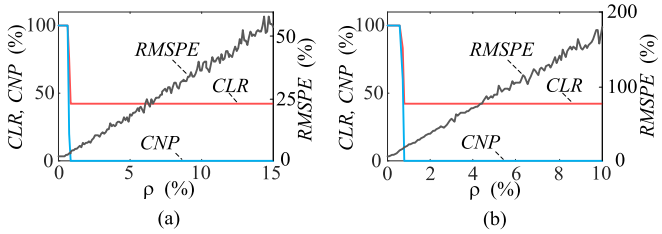


Fig. 10. Results of the Bayesian method (IEEE 13-bus system, case B2). (a) 5th harmonic, Bayesian method and (b) 13th harmonic, Bayesian method.

To further verify the validity of the proposed method, the number of harmonic sources is increased to 3, as reported in case B2 of Table XII. In this case, the average computational time of the proposed method is 0.02 s.

Table XII and Figs. 9 and 10 present the corresponding estimated results. In Table XII, the CLRs estimated by the proposed method are slightly higher than those from the BOMP method for the fifth and 11th harmonics. However, for the fifth harmonic, the RMSPE estimated by the BOMP method is up to 40% resulted from the outliers with large APEs, while, for the proposed method, the maximum RMSPE for all harmonic orders is below 7%, and the estimated APEs in Fig. 9(b) are concentrated at 3% for the seventh and 11th harmonics. More importantly, the proposed method needs less prior information on suspicious harmonic sources than the compared methods.

Fig. 10 shows the estimated results of the Bayesian method. The trends of these curves are similar to those in case A1, which indicates that the effectiveness of the Bayesian method is dominated by the prior information of suspicious harmonic sources.

V. FIELD CASE VERIFICATIONS

The 220-kV interconnected transmission network is used to verify the effectiveness of the proposed method. As shown in Fig. 11, this network contains five single transmission lines from substation YX to TM, DW, JT, JS, and YL and one double transmission lines between substation LW and YX. The parameters of the transformers, lines, and loads of the fundamental frequency are provided by State Grid Corporation of China.

The transmission lines are modeled as a single-phase equivalent PI model with distributed parameters [40], [45]. The loads are modeled as impedances based on the active and reactive powers of the fundamental frequency [12], [45], [46]. Set $N = 20$ and $M = 30$.

This test system is an overdetermined measurement system with a total of 18 HMs synchronized with a GPS clock.

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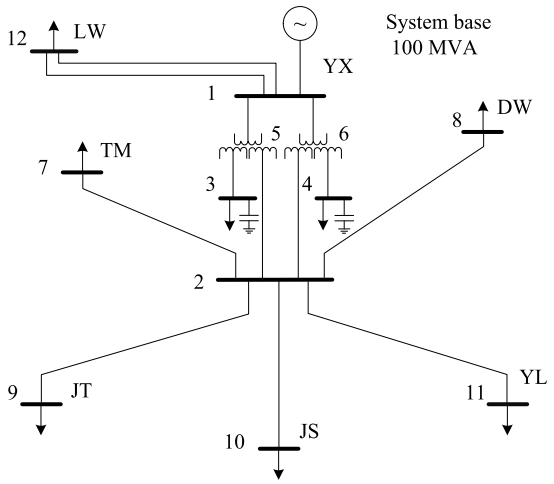


Fig. 11. Field test system.



Fig. 12. Experimental platform for the harmonic source estimation.

The HMs measuring voltage phasors are on buses 1, 2, 3, 4, 10, and 11, and the HM's measuring current phasors are on the branches 1–5, 1–6, 2–5, 2–6, 3–5, 4–6, 1–12, 2–7, 2–8, 2–9, 2–10, and 2–11. The sampling rate of the measurements is 51.2 kHz (50-Hz system), and the measurement accuracy meets the accuracy requirements of the Class I instrument in standard Chinese GB/T 17626.7-2017. The experimental platform is shown in Fig. 12, where the voltage/current of measured buses/lines is transformed and measured by the electronic transformers and measurement devices; meanwhile, the discrete Fourier transformation is carried out on the measured data to obtain the harmonic data, and the fifth and seventh harmonic data are shown in Fig. 13.

Since the harmonic current emissions of loads cannot be directly measured, in order to evaluate the reliability of the proposed method, the estimated current emissions derived from the overdetermined measurement equations are used as the reference values, which can be solved by the least-squares method with using the measured data of all the 18 available HMs. In the estimated results (at the 30th hour), buses 11 and 12 have extremely high emission levels of the fifth and seventh harmonic currents than other buses. Thus, buses 11 and 12 are considered to contain harmonic sources.

Then, the measured fifth and seventh harmonic data in Fig. 13 are used to locate and estimate the harmonic sources. For testing the effectiveness of the proposed method

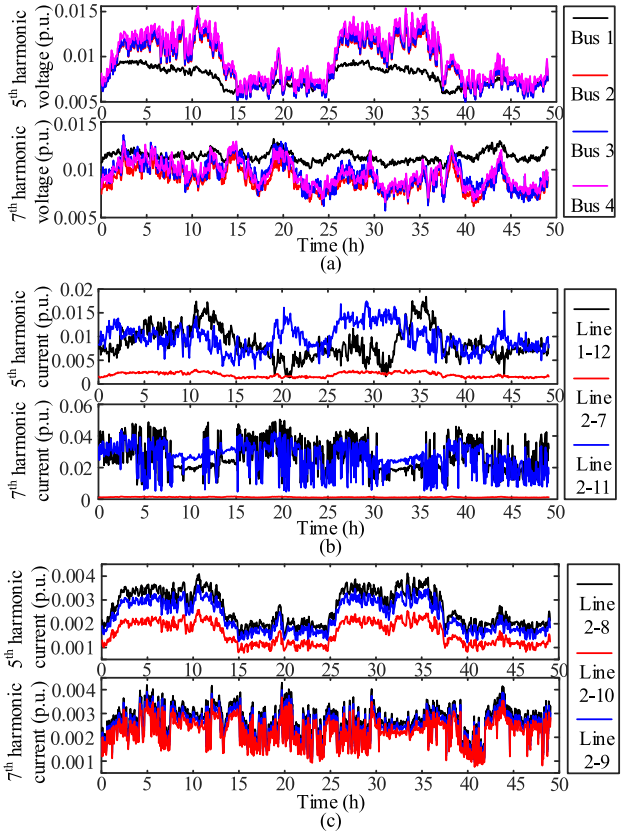


Fig. 13. Measured harmonic data: (a) fifth and seventh harmonic voltages and (b) and (c) fifth and seventh harmonic currents.

TABLE XIII
HMs' PLACEMENTS

Field case	Meter placement
Field case 1	bus 1, 2, 3, 4, line 1-12, 2-7, 2-11
Field case 2	bus 1, 2, 3, 4, line 2-8, 2-9, 2-10
Field case 3	bus 1, 2, line 2-7, 2-8, 2-9, 2-11
Field case 4	bus 1, 3, 4, line 2-7, 2-8, 2-10

TABLE XIV
MINIMUM FITTING RESIDUALS

Field case	Harmonic order	$r_{\min}(k)$		
		k=1	k=2	k=3
Field case 1	5th	0.0082	0.0006	0.0004
	7th	0.0445	0.0025	0.0016
Field case 2	5th	0.0057	0.0003	0.0001
	7th	0.0373	0.0017	0.0015
Field case 3	5th	0.0059	0.0004	0.0002
	7th	0.0293	0.0012	0.0008
Field case 4	5th	0.0023	0.0004	0.0002
	7th	0.0248	0.0009	0.0007

under different placements of HMs, the four field cases shown in Table XIII are considered, and each case is the underdetermined case with six or seven HMs randomly selected from all available measurements.

The estimated results are shown in Tables XIV–XVI. In Table XIV, the minimum fitting residuals for calculating the number of harmonic sources are presented. It can be seen

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TABLE XV
RESULTS OF LOCATED HARMONIC SOURCES

Field case	Harmonic order	Buses of Located harmonic sources
Field case 1	5th	11,12
	7th	11,12
Field case 2	5th	11,12
	7th	11,12
Field case 3	5th	11,12
	7th	11,12
Field case 4	5th	11,12
	7th	11,12

TABLE XVI
RESULTS OF APES

Field case	Harmonic order	*APE*
Field case 1	5th	12.9%
	7th	6.9%
Field case 2	5th	4.5%
	7th	7.4%
Field case 3	5th	5.9%
	7th	3.3%
Field case 4	5th	9.8%
	7th	4.7%

that, for each analyzed case, when k changes from 2 to 3, the variation of the minimum fitting residual becomes smaller than $0.05r_{\min}(1)$, and thus, the number of harmonic sources is determined as two that correctly reflects the true number of harmonic sources.

The results of harmonic source location and estimation are, respectively, reported in Tables XV and XVI. As from the data in the two tables, in each considered case, the proposed method can accurately locate harmonic sources of the fifth and seventh harmonics, and the largest APEs for the fifth and seventh harmonics are, respectively, 12.9% and 7.4%. The appeal analysis further proves the effectiveness of the proposed method under different placements of HMs.

VI. CONCLUSION

This article proposes a novel method for harmonic source location and estimation, while the number of synchronized HMs is less than that of unknown state variables. The effectiveness of the proposed method is verified by both simulation cases and field cases. In particular, the performances of the methods under the different placement of HMs are further discussed. Furthermore, the uncertainties of the estimation results caused by measurement errors and parameter errors are analyzed by means of the Monte Carlo method.

The advantages of the proposed method can be summarized as: 1) it needs less prior information of harmonic sources; 2) the validity of the method is independent of the placement of HMs; and 3) it has a higher accuracy for locating harmonic sources compared with the existing methods formulated for the underdetermined measurement system.

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